Projects for Lab Examination 1

Note:

1. Implement the project below. You are allowed to use AI-based or other tools for this project (if you wish). Evaluation at the time of the examination will be carried out for each student separately.
2. You must write a brief description of your design of the project and add meaningful comments in your code.
3. This implementation carries no marks on its own. You will be asked to modify your own code for the project during the examination to add more features, and those changes will carry 100% marks. Different students may get different features to implement during the examination. **No AI-based or other tools will be allowed during the examination.** All the modifications must be done on your own.
4. You must submit the implementation (design explanation+code+test cases) to **Mr. Nikhil S., latest by 11 PM on August 26, 2026**. He will tell you the method of submission in the lab class. The first lab exam will be scheduled on **August 28, 2026, 9 AM-11 AM** for both sections of CS.

Projects

1. Implement a simulator for a small and simple subset of Java byte code in C. Assume maximum sizes of the operand stack and register set (local variables) to be 10 each (you may use a larger size, if you need it). The subset of the instruction set that you need to simulate consists of **{ldc, iload, istore, iadd, isub, imul, idiv, ifeq, iflt, ifgt},** and two special instructions, **{read, print},** that are not available in Java byte code. **{read}** reads an integer from the keyboard, and pushes it onto the operand stack. Similarly, **{print}** prints the integer on the top of the stack on the console. You need to implement the **{ldc}** instruction for integers only.

   The input to the simulator is a Java byte code program in ordinary text format, with one instruction per line. Read the instructions from the input and store them suitably before beginning the simulation. Supply a set of non-trivial programs in this subset of Java byte code that you have tested the simulator with, along with Java-like high level equivalent code for the test programs. You may learn details of the byte codes from the **URL:https://docs.oracle.com/javase/specs/jvms/se11/html/jvms-6.html**

   You must supply THREE test programs for the following tasks:

   a. Read three integers from the input, store them, find their minimum and maximum, store them, and print them out.
   b. Read an integer *n* from the input, then read *n* integers from the input, sort them in ascending order, and then print them out.
   c. Read two 2x2 matrices from the input. You may read the numbers in the order, a_{11}, a_{12}, a_{21}, and a_{22}. Compute their sum and print out the sum in the same order described above.